Business Requirements Document (BRD)

Project: Open-Core Intern Management System (IMS) — "The HR Ticket Translator" Doc Ref: IMS-BRD-1.0

1. Executive Summary & Business Purpose

Problem Statement: Organizations running internship and junior training programs suffer from a systemic "Swivel Chair" disconnect between two incompatible software categories. Execution trackers (Jira, Linear, Kanboard) measure shipping velocity but are blind to intern learning curves, skill gaps, and evaluation rubrics. Conversely, corporate HR platforms (BambooHR, Lattice, Frappe HR, GoSprout) enforce appraisal compliance but are disconnected from daily task work, forcing mentors to rely on subjective end-of-month memory.

Product Vision: The IMS bridges this divide by operating as an automated **Ticket Translator**. It gives mentors a lightweight, continuous-flow Kanban board while silently running an immutable **Transition Ledger** and an event-driven **"Justify This Action" Intercept Engine**. Everyday board movements are automatically translated into objective, timestamped competency metrics and injected directly into HR's 30/60/90-day evaluation rubrics.

2. Market Positioning & Go-To-Market (GTM) Strategy

Market Tier	Target Segment	Org Size	Adoption Motion & Strategic Rationale
Primary Wedge (Open-Core MVP)	Mid-Market Tech, Dev Shops & Digital Agencies	50 – 500 Employees	**Bottom-Up (PLG):** Recurring intern cohorts; DevOps teams readily self-host Dockerized tools without HR procurement delays.
Horizontal Expansion	Finance, Marketing, Accounting & Schools	100 – 1,000 Employees	**Department-Agnostic Flow:** Custom Kanban states adapt seamlessly to financial models, copywriting, and lesson plans.
Enterprise Tier (ee/ Commercial)	FinTech, Banking, Healthcare IT & Regulated Corps	500+ Employees	**Top-Down Security Upgrade:** CISOs mandate paid licenses for SAML/SSO, SCIM provisioning, PBAC, and SIEM audit streams.

Explicit Strategic Non-Goals (Out of Scope):
- **Government Blue-Collar Apprenticeships:** No Department of Labor grant tracking or trade credit-hour compliance (ceded to GoSprout).
- **Client Billing Stopwatches:** Tracks passive state-duration friction for learning velocity, not billable client invoicing (ceded to Harvest).
- **Rigid Scrum Ceremonies:** Zero sprint burndown charts, story-point poker, or Scrum Master role overhead; strictly continuous-flow Kanban.

3. Stakeholder & User Personas

The system serves four distinct organizational personas, aligning daily engineering convenience with executive HR governance:

1. The Champion — Mentor / Tech Lead

Objective: Assign tasks, review deliverables, and guide interns without filling out Friday HR spreadsheets.

Requirement: Fast drag-and-drop board; ability to log performance context in <5 seconds when kicking a task back.

2. The End User — The Intern / Mentee

Objective: Understand expectations, ship real work, receive timely feedback, and earn a full-time return offer.

Requirement: Fair protection when tasks stall due to external blockers (IT access, client delays, or slow mentor reviews).

3. Economic Buyer — HR / People Ops

Objective: Standardize evaluations across cohorts, spot struggling interns at Day 30, and keep a legal paper trail.

Requirement: Automated 30/60/90-day forms pre-populated with objective revision counts and categorized skill gaps.

4. The Gatekeeper — DevOps / IT Admin

Objective: Deploy, secure, and maintain internal software with minimal server cost and zero dependency churn.

Requirement: Single compiled Go binary, low RAM footprint, clean Docker Compose setup, and offline JWT licensing.

4. Functional Business Requirements

4.1 MODULE 1: DEPARTMENT-AGNOSTIC WORKFLOW ENGINE (OPEN-CORE — PRIORITY P0)

- `BR-WF-01` **Custom State Machines:** Department admins must be able to define custom Kanban columns in any sequence (e.g., *"Drafting"* → *"Awaiting MD Approval"* → *"Client Feedback"* → *"Finalized"*) without hardcoded software terminology.
- `BR-WF-02` **Mandatory Skill Tagging:** Every task must carry at least one department-defined competency tag (e.g., `#Python`, `#DCF-Valuation`, `#Copywriting`) so workflow friction maps directly to human skills.
- `BR-WF-03` **Append-Only Transition Ledger:** Task state history must never be overwritten. Every card movement appends an immutable row logging `task_id`, `from_column_id`, `to_column_id`, `actor_id`, timestamp, and `time_spent_in_previous_ms`.
- `BR-WF-04` **Configurable SLA Thresholds & Neutral Nudges:** Admins may attach an optional `max_hours_in_state` limit to any column. When breached, a background worker dispatches a neutral alert to the mentor without assuming fault.

4.2 Module 2: The "Justify This Action" Intercept Engine (Open-Core — P0)

- `BR-INT-01` **Deterministic Anomaly Triggers:** The state machine must intercept card movements under three conditions: (1) **Revision Bounce-Back** (card moves backward to an earlier column for the ≥2nd time), (2) **Severe Stagnation** (time in column exceeds 300% of SLA threshold), or (3) **Blocked Oscillation** (card enters a Blocked state >2 times in 7 days).
- `BR-INT-02` **Synchronous Precondition Modal:** When triggered, the backend halts the transition (`HTTP 428 Precondition Required`) and prompts the mentor to classify the root cause before the card drop commits.
- `BR-INT-03` **Standardized Fault & Attribution Taxonomy:** Mentors select from four mutually exclusive classifications that govern downstream HR calculations:

Justification Code	Root Cause Definition	Fault Attribution	Automated HR Rubric Impact
`KNOWLEDGE_GAP`	Intern lacks domain or technical understanding required for the deliverable.	**Intern (Skill)**	Flags the task's `#Skill_Tag` for targeted pairing; logs competency gap evidence.
`CARELESS_ERROR`	Intern understands the material but submitted rushed or unverified work.	**Intern (Habit)**	Flags the "Attention to Detail & Quality" behavioral rubric dimension.
`SCOPE_CREEP`	Requirements shifted mid-task or deliverable was poorly scoped initially.	**Mentor / Product**	**Protects Intern:** Excludes revision loop and overrun duration from intern metrics.
`THIRD_PARTY_BLOCKER`	Delayed by external dependency (IT permissions, client review, vendor).	**External / System**	**Protects Intern:** Strips blocked hours from cycle time; logs org bottleneck.

4.3 Module 3: HR Translation, Rubrics & Cohort Lifecycles (Open-Core — P0/P1)

- `BR-HR-01` **Cohort Lifecycle Management:** Group interns into time-bound cohorts (e.g., *"Summer 2027"*) with automated review checkpoints at Day 30, Day 60, and Day 90.
- `BR-HR-02` **Dynamic Rubric Builder:** Allow HR admins to build department-specific evaluation forms with 1–5 Likert scales, weighted criteria, and self-appraisals stored flexibly via PostgreSQL `JSONB`.
- `BR-HR-03` **Automated Telemetry Injection:** Pre-populate 30/60/90-day evaluation forms with an uneditable *Execution Evidence Summary* (tasks completed per skill, cycle time vs. cohort average, and intercept breakdown).
- `BR-HR-04` **Structured 1-on-1 Syncs & Hiring Matrix:** Provide a shared weekly 1-on-1 notepad (public agenda + private mentor notes) culminating in a final **Recommend for Return Offer [Yes / No]** workflow.

4.4 Module 4: Enterprise Governance & Monetization (ee/ Tier — Priority P2)

- `BR-EE-01` **Offline Cryptographic Licensing (RS256):** Gate all modules inside `ee/` behind an air-gapped RS256 asymmetric JWT license check validating organization domain, active intern seat quota, and expiration against an embedded RSA public key.
- `BR-EE-02` **Enterprise SSO & SCIM Provisioning:** Support SAML 2.0 / OIDC authentication (Okta, Microsoft Entra ID) and automated SCIM 2.0 user and cohort lifecycle provisioning.
- `BR-EE-03` **Policy-Based Access Control (PBAC):** Enable granular permission policies isolating cohort data across global branches, departments, and HR business partners.
- `BR-EE-04` **SIEM Audit Streaming:** Stream immutable administrative and evaluation events in real time to enterprise security platforms (Datadog, Splunk, Elastic).

5. Non-Functional, Solo-Founder & Legal Requirements

- `NFR-01` **Single-Binary Operational Simplicity:** Compile the SvelteKit UI (`@sveltejs/adapter-static`) directly into the Go binary via `//go:embed` so the entire platform ships as one executable alongside PostgreSQL and Valkey.
- `NFR-02` **Hybrid Relational + JSONB Persistence:** Enforce strict relational schemas with `SQLc` for the Transition Ledger and Intercepts, while using PostgreSQL `JSONB` mapped to Go structs (`pgx/v5`) for dynamic HR forms to eliminate CRUD boilerplate.
- `NFR-03` **Permissive Dependency Guarantee:** Restrict all dependencies to permissive licenses (MIT, BSD-3, Apache 2.0), utilizing **Valkey** (BSD-3) with Asynq to ensure zero copyleft contamination of the commercial binary.
- `NFR-04` **GDPR Personnel File Anonymization:** Provide an automated retention job (configurable 1–6 years post-cohort) that scrubs PII (`intern_id`, names, free-text notes) while preserving anonymized transition timings for department benchmarks.
- `NFR-05` **Dual-License & CLA Enforcement:** Govern core directories under MIT, isolate `ee/` under a commercial EULA prohibiting JWT bypass, and enforce automated GitHub CLA signing on all external pull requests.

6. Business Success Metrics & Target KPIs

Metric Category	Key Performance Indicator (KPI)	Target Benchmark
Developer Adoption	Time from `docker compose up` to first active department Kanban board	**< 5 minutes**
Friction Attribution	Percentage of backward task bounces captured with a structured Justification Code	**100%** (HTTP 428 enforced)
Mentor Time Savings	Average time required for a mentor to complete a 30-day intern appraisal	**< 4 minutes** (down from ~30m)
Enterprise Conversion	Self-hosted community organizations converting to paid `ee/` subscriptions	**3% – 5%** conversion rate